

**5.0.1 NIELIT 2017 July Scientist B (CS) - Section B: 26**

The process of converting the analog sample into discrete form is called

- A. Modulation B. Multiplexing C. Quantization D. Sampling

nielit2017july-scientistb-cs computer-networks

Answer key

5.0.2 NIELIT 2016 DEC Scientist B (CS) - Section B: 20

Bluetooth is an example of:

- A. Personal area network B. Virtual private network
C. Local area network D. None of the above

nielit2016dec-scientistb-cs computer-networks

Answer key

5.0.3 NIELIT 2016 DEC Scientist B (CS) - Section B: 19

Communication between a computer and a keyboard involves _____ transmission

- A. Simplex B. Half-Duplex C. Automatic D. Full-Duplex

nielit2016dec-scientistb-cs computer-networks

Answer key

5.0.4 NIELIT 2016 DEC Scientist B (CS) - Section B: 9

The first Network:

- A. ARPANET. B. NFSNET. C. CNET. D. ASAPNET.

nielit2016dec-scientistb-cs computer-networks

Answer key

5.0.5 NIELIT 2016 DEC Scientist B (IT) - Section B: 54

Which of this is not a network edge device?

- A. PC B. Server C. Smartphone D. Switch

nielit2016dec-scientistb-it computer-networks

Answer key

5.0.6 NIELIT 2016 DEC Scientist B (IT) - Section B: 22



Bluetooth is an example of:

- A. Local area network.
- B. Virtual private network.
- C. Personal area network.
- D. None of the mentioned above

nielit2016dec-scientistb-it computer-networks

Answer key

5.0.7 NIELIT 2016 DEC Scientist B (IT) - Section B: 5



Two alternative packages A and B are available for processing a database having $10k$ records. Package A requires $0.0001n^2$ time units and package B requires $10n \log 10n$ time units to process n records. What is the smallest value of k for which package B will be preferred over A ?

- A. 12
- B. 10
- C. 6
- D. 5

nielit2016dec-scientistb-it computer-networks

Answer key

5.0.8 NIELIT 2016 DEC Scientist B (IT) - Section B: 2



Three or more devices share a link in _____ connection

- A. Unipoint
- B. Polarpoint
- C. Point to point
- D. Multipoint

nielit2016dec-scientistb-it computer-networks

Answer key

5.0.9 NIELIT 2017 OCT Scientific Assistant A (CS) - Section B: 26



If the channel is band limited to 6 kHz and signal to noise ratio is 16, what would be the capacity of channel?

- A. 16.15 kbps
- B. 23.24 kbps
- C. 40.12 kbps
- D. 24.74 kbps

nielit2017oct-assistanta-cs computer-networks

Answer key

5.0.10 NIELIT 2017 OCT Scientific Assistant A (IT) - Section B: 32



The maximum data rate of a channel of 3000-Hz bandwidth and SNR of 30 db is

- A. 60000
- B. 15000
- C. 30000
- D. 3000

nielit2017oct-assistanta-it computer-networks

Answer key

5.0.11 NIELIT 2016 MAR Scientist C - Section C: 33



In networking terminology UTP means

- A. Unshielded Twisted pair
- B. Ubiquitous Teflon port
- C. Uniformly Terminating port
- D. Unshielded T- connector port

nielit2016mar-scientistc computer-networks

Answer key

5.0.12 NIELIT 2016 MAR Scientist C - Section C: 22



What are the primary characteristics that distinguish a cell from a packet?

- A. cells are generally smaller than packets
- B. cells do not incorporate physical address
- C. all cells have the same fixed length
- D. packet cannot be switched

nielit2016mar-scientistc computer-networks

Answer key

5.0.13 NIELIT 2016 MAR Scientist C - Section C: 21



Repeaters function in

- A. Physical layer
- B. Data link layer
- C. Network layer
- D. Both (A) and (B)

nielit2016mar-scientistc computer-networks

Answer key

5.0.14 NIELIT 2016 MAR Scientist C - Section C: 20



Start and stop bits are used in serial communication for

- A. error detection
- B. error correction
- C. synchronization
- D. slowing down the communication

nielit2016mar-scientistc computer-networks

Answer key

5.0.15 NIELIT Scientific Assistant A 2020 November: 82



To guarantee correction of upto 5 errors in all cases, the minimum Hamming distance in a block code must be _____.

- A. 11
- B. 6
- C. 5
- D. 2

nielit-sta-2020 computer-networks hamming-code

Answer key

5.0.16 NIELIT Scientific Assistant A 2020 November: 43



In ICMP, in case of time exceeded error, when the datagram visits a router, the value of time to live field is _____

- A. Remains constant
- C. Incremented by 1

- B. Decrement by 2
- D. Decrement by 1

nielit-sta-2020 computer-networks icmp

Answer key

5.0.17 NIELIT Scientist B 2020 November: 88



In which modulation discrete values of carrier frequencies is used to transmit binary data?

- A. Phase Shift Keying
- B. Amplitude Shift Keying
- C. Frequency Shift Keying
- D. Disk Shift Keying

nielit-scb-2020 computer-networks

Answer key

5.0.18 NIELIT Scientist B 2020 November: 83



_____ possible labels are allowed in the first level of generic domain.

- A. 10
- B. 12
- C. 16
- D. none of the options

nielit-scb-2020 computer-networks

Answer key

5.0.19 NIELIT 2021 Dec Scientist B - Section B: 93



The spectral efficiency of QPSK null to null having a width 4 Hz will be :

- A. 1/2
- B. 1
- C. 1/4
- D. 4

nielit2021dec-scientistb computer-networks

5.0.20 NIELIT 2021 Dec Scientist B - Section B: 59



A wireless network interface controller can work in _____ .

- A. infrastructure mode
- B. ad-hoc mode
- C. both infrastructure mode and ad-hoc mode
- D. WDS mode

nielit2021dec-scientistb computer-networks

Answer key

5.1 Application Layer (1)

5.1.1 Application Layer: NIELIT 2016 DEC Scientist B (IT) - Section B: 49



Application layer protocol defines:

- A. types of messages exchanged
- B. rules for when and how processes send and respond to messages
- C. message format, syntax and semantics
- D. all of the above

Answer key 

5.2 Application Layer Protocols (2)

5.2.1 Application Layer Protocols: NIELIT 2021 Dec Scientist A - Section B: 44

The File Transfer Protocol is built on _____.



- | | |
|-------------------------------|-------------------------------------|
| A. data centric architecture | B. service-oriented architecture |
| C. client server architecture | D. connection-oriented architecture |

nielit2021dec-scientista computer-networks application-layer-protocols

Answer key 

5.2.2 Application Layer Protocols: NIELIT 2021 Dec Scientist B - Section B: 28

Which TCP/IP protocol is used for file transfer with minimal capability and minimal overhead?



- | | | | |
|---------|--------|---------|-----------|
| A. RARP | B. FTP | C. TFTP | D. TELNET |
|---------|--------|---------|-----------|

nielit-2021-it-dec-scientistb computer-networks transport-layer application-layer-protocols

Answer key 

5.3 Baud Rate (1)

5.3.1 Baud Rate: NIELIT 2018-35

Baud rate measures the number of _____ transmitted per second



- | | | | |
|------------|---------|---------|------------------|
| A. Symbols | B. Bits | C. Byte | D. None of these |
|------------|---------|---------|------------------|

nielit-2018 computer-networks baud-rate

Answer key 

5.4 Bit Rate (1)

5.4.1 Bit Rate: NIELIT 2022 April Scientist B | Section B | Question: 60

For a bit-rate of 8 kbps, the best possible values of the transmitted frequencies in a coherent binary FSK system are :



- | | |
|----------------------|----------------------|
| A. 16 kHz and 20 kHz | B. 20 kHz and 32 kHz |
| C. 20 kHz and 40 kHz | D. 32 kHz and 40 kHz |

nielit2022apr-scientistb computer-networks data-communication modulation bit-rate

5.5 Bit Stuffing (2)

5.5.1 Bit Stuffing: NIELIT 2017 OCT Scientific Assistant A (CS) - Section B: 24

Bit stuffing refers to



- A. Inserting a '0' in user data stream to differentiate it with a flag
- B. Inserting a '0' in flag stream to avoid ambiguity
- C. Appending a nibble to the flag sequence
- D. Appending a nibble to the user data stream

nielit2017oct-assistanta-cs computer-networks bit-stuffing

Answer key 

5.5.2 Bit Stuffing: NIELIT 2017 OCT Scientific Assistant A (IT) - Section C: 4



Bit stuffing refers to

- A. Inserting a '0' in user data stream to differentiate it with a flag
- B. Inserting a '0' in flag stream to avoid ambiguity
- C. Appending a nibble to the flag sequence
- D. Appending a nibble to the user data stream

nielit2017oct-assistanta-it computer-networks bit-stuffing

Answer key 

5.6 CO and Architecture (1)

5.6.1 CO and Architecture: NIELIT 2021 Dec Scientist B - Section B: 68



What is the minimum no. of wires required for sending data over a serial communication link?

- A. 1 B. 3 C. 2 D. 4

nielit-2021-it-dec-scientistb computer-networks co-and-architecture

5.7 CRC Polynomial (3)

5.7.1 CRC Polynomial: NIELIT 2018-79



Given message $M = 1010001101$. The CRC for this given message using the divisor polynomial $x^5 + x^4 + x^2 + 1$ is _____

- A. 01011 B. 10101 C. 01110 D. 10110

nielit-2018 computer-networks crc-polynomial

Answer key 

5.7.2 CRC Polynomial: NIELIT 2021 Dec Scientist A - Section B: 115



If the CRC has Polynomial of degree n , then what is the probability of detecting errors greater than n ?

- A. $\frac{1}{2^{n-1}}$ B. $\frac{1}{2^{n+1}}$ C. $\frac{1}{2^n}$ D. $\frac{1}{2^{n+2}}$

nielit2021dec-scientista crc-polynomial probability error-detection computer-networks

5.7.3 CRC Polynomial: NIELIT Scientific Assistant A 2020 November: 120



In CRC calculation if divisor is 1011, and dataword is 1001 what will be the CRC?

- A. 111 B. 101 C. 110 D. 100

nielit-sta-2020 computer-networks crc-polynomial

Answer key

5.8

CSMA CD (1)

5.8.1 CSMA CD: NIELIT 2021 Dec Scientist A - Section B: 73



In CSMA/CD after detecting the collision, station immediately stops transmission by sending the _____.

- A. Stop pattern B. Preamble pattern
C. Jam signal D. Block signal

nielit2021dec-scientista computer-networks csma-cd

Answer key

5.9

CSMA Ca (2)

5.9.1 CSMA Ca: NIELIT 2017 DEC Scientific Assistant A - Section B: 13



Which multiple access technique is used by IEEE 802.11 standard for wireless LAN ?

- A. CDMA B. CSMA/CA C. ALOHA D. None of the options

nielit2017dec-assistanta computer-networks csma-ca

Answer key

5.9.2 CSMA Ca: NIELIT 2021 Dec Scientist A - Section B: 112



Which multiple access technique is used by IEEE 802.11 standard for wireless LAN ?

- A. CDMA B. CSMA/CA C. ALOHA D. CSMA/CD

nielit2021dec-scientista computer-networks csma-cd csma-ca lan-technologies

Answer key 

5.10

Channel Capacity (1)

5.10.1 Channel Capacity: NIELIT 2017 OCT Scientific Assistant A (IT) - Section C: 11

If the channel is band limited to 6 kHz and signal to noise ratio is 16, what would be the capacity of channel?



- A. 16.15 kbps B. 23.24 kbps C. 40.12 kbps D. 24.74 kbps

nielit2017oct-assistanta-it computer-networks channel-capacity

Answer key 

5.11

Communication (2)

5.11.1 Communication: NIELIT 2021 Dec Scientist A - Section B: 46

In DPSK technique, the technique used to encode bits is :



- A. AMI B. Differential code
C. Unipolar RZ format D. Manchester format

nielit2021dec-scientista communication

Answer key 

5.11.2 Communication: NIELIT 2022 April Scientist B | Section B | Question: 119

In a binary data transmission DPSK is preferred to PSK because :



- A. a coherent carrier is not required to be generated at the receiver
B. for a given energy per bit, the probability of error is less
C. the 1800 phase shifts of the carrier are unimportant
D. more protection is provided against impulse noise

nielit2022apr-scientistb computer-networks communication

Answer key 

5.12

Congestion Control (4)

5.12.1 Congestion Control: NIELIT 2017 DEC Scientific Assistant A - Section B: 11

Closed - Loop control mechanism try to :



- A. Remove congestion after it occurs B. Remove congestion after sometime
C. Prevent congestion before it occurs D. Prevent congestion before sending packets

nielit2017dec-assistanta computer-networks congestion-control

Answer key 

5.12.2 Congestion Control: NIELIT 2017 DEC Scientific Assistant A - Section B: 43

When we use slow-start algorithm, the size of the congestion window increases _____ until it reaches a threshold.



- A. Additively
- B. Multiplicatively
- C. Exponentially
- D. None of the options

nielit2017dec-assistanta computer-networks congestion-control sliding-window

Answer key 

5.12.3 Congestion Control: NIELIT Scientific Assistant A 2020 November: 87



Why does congestion occur?

- A. Because the routers and switches have tables
- B. Because the routers and switches have queues
- C. Because the routers and switches have cross-points
- D. None of the options

nielit-sta-2020 computer-networks congestion-control

Answer key 

5.12.4 Congestion Control: NIELIT Scientist B 2020 November: 98



In the congestion avoidance algorithm, the size of the congestion window increases _____ until congestion is detected.

- A. Exponentially
- B. Additively
- C. Multiplicatively
- D. Suddenly

nielit-scb-2020 computer-networks congestion-control

Answer key 

5.13 Cryptography (1)

5.13.1 Cryptography: NIELIT Scientist B 2020 November: 78



_____ uses pretty good privacy algorithm.

- A. Electronic mails
- B. File encryption
- C. Both Electronic mails and File encryption
- D. None of the options

nielit-scb-2020 network-security cryptography computer-networks

Answer key 

5.14 Data Communication (2)

5.14.1 Data Communication: NIELIT 2021 Dec Scientist B - Section B: 86



The bit rate of digital communication system is R kbit/s. The modulation used is 32-QAM. The minimum bandwidth required for ISI free transmission is :

- A. $R/10\text{ Hz}$
- C. $R/5\text{ Hz}$

- B. $R/10\text{ kHz}$
- D. $R/5\text{ kHz}$

nielit2021dec-scientistb data-communication digital-logic

5.14.2 Data Communication: NIELIT Scientist B 2020 November: 114



A special *PCM* system uses 32 channels of data, one whose purpose is an identification (ID) and synchronization. The sampling rate is 4 kHz. The word length is 5 bits. Find the serial data rate.

- A. 1280 kHz
- B. 160 kHz
- C. 320 kHz
- D. 640 kHz

nielit-scb-2020 co-and-architecture data-communication

Answer key

5.15 Data Path (1)

5.15.1 Data Path: NIELIT 2016 DEC Scientist B (IT) - Section B: 47



The _____ is the physical path over which a message travels.

- A. Path
- B. Protocol
- C. Route
- D. Medium

nielit2016dec-scientistb-it computer-networks data-path

Answer key

5.16 Digital Signal Processing (1)

5.16.1 Digital Signal Processing: NIELIT 2021 Dec Scientist A - Section B: 65



Equalization process includes :

- A. maximum likelihood sequence estimation and equalization with filters
- B. maximum likelihood sequence estimation
- C. equalization with filters
- D. constant impulse response

nielit2021dec-scientista computer-networks digital-signal-processing

5.17 Digital Signature (1)

5.17.1 Digital Signature: NIELIT Scientific Assistant A 2020 November: 62



A digital signature is required:

- A. for non-repudiation of communication by a sender
- B. for all e-mail sending
- C. for all DHCP server
- D. for FTP transactions

Answer key 

5.18

Error Correction (1)

5.18.1 Error Correction: NIELIT 2017 OCT Scientific Assistant A (CS) - Section B: 2

Which of the following is minimum error code?



- A. Octal Code B. Binary Code C. Gray Code D. Excess-3 Code

nielit2017oct-assistanta-cs computer-networks error-correction

Answer key 

5.19

Error Detection (2)

5.19.1 Error Detection: NIELIT 2017 OCT Scientific Assistant A (CS) - Section B: 22

If the original size of data is 40 then after adding error detection redundancy bit the size of data length is



- A. 26 B. 36 C. 46 D. 56

nielit2017oct-assistanta-cs computer-networks error-detection

Answer key 

5.19.2 Error Detection: NIELIT Scientist B 2020 November: 60

To the detection of up to 5 errors in all cases, the minimum Hamming distance in a block code must be _____.



- A. 5 B. 6 C. 10 D. 8

nielit-scb-2020 error-detection digital-logic

Answer key 

5.20

Fragmentation (3)

5.20.1 Fragmentation: NIELIT 2017 DEC Scientific Assistant A - Section B: 51

The demerits of the fragmentation are :



- A. Complex routers B. Open to DOS attack
C. No overlapping of fragments D. (A) and (B) both

nielit2017dec-assistanta computer-networks fragmentation

Answer key 

5.20.2 Fragmentation: NIELIT 2022 April Scientist B | Section B | Question: 75

In an IPv4 datagram, the M bit is 0, the value of HLEN is 10, the value of total



length is 400 and the fragment offset value is 300. The position of the datagram, the sequence numbers of the first and the last bytes of the payload, respectively are:

- A. Last fragment, 2400 and 2789
- B. First fragment, 2400 and 2759
- C. Last fragment, 2400 and 2759
- D. Middle fragment, 300 and 689

nielit2022apr-scientistb computer-networks ip-addressing fragmentation

5.20.3 Fragmentation: NIELIT Scientific Assistant A 2020 November: 75



The _____ field in IPv4 datagram is not related to fragmentation.

- A. Flag
- B. Offset
- C. TOS
- D. Identifier

nielit-sta-2020 computer-networks fragmentation

Answer key

5.21 General Awareness (1)

5.21.1 General Awareness: NIELIT 2021 Dec Scientist B - Section B: 76



SPAM usually _____.

- A. Contains no valuable content.
- B. Contains valuable content.
- C. Contains content about website and is an operating system.
- D. Is a module of operating system that handles scheduling of a process.

nielit-2021-it-dec-scientistb general-awareness analytical-aptitude

5.22 IP Addressing (7)

5.22.1 IP Addressing: NIELIT 2016 DEC Scientist B (IT) - Section B: 19



Which protocol finds the MAC address from IP address?

- A. SMTP
- B. ICMP
- C. ARP
- D. RARP

nielit2016dec-scientistb-it computer-networks ip-addressing

Answer key

5.22.2 IP Addressing: NIELIT 2017 DEC Scientific Assistant A - Section B: 12



In classless addressing, there are no classes but addresses are still granted in :

- A. Codes
- B. Blocks
- C. IPs
- D. Sizes

nielit2017dec-assistanta computer-networks ip-addressing subnetting

Answer key

5.22.3 IP Addressing: NIELIT 2017 DEC Scientific Assistant A - Section B: 24



A subnet mask in class *C* can have _____ 1's with the remaining bits 0's.

- A. 10 B. 24 C. 12 D. 7

nielit2017dec-assistanta computer-networks subnetting ip-addressing

Answer key

5.22.4 IP Addressing: NIELIT 2017 DEC Scientific Assistant A - Section B: 45



In *IPv4* Addresses, classfull addressing is replaced with :

- A. Classless Addressing B. Classfull Addressing
C. Subnet Advertising D. None of the options

nielit2017dec-assistanta computer-networks ip-addressing ip-addressing

Answer key

5.22.5 IP Addressing: NIELIT 2018-68



_____ IP address can be used in WAN

- A. 256.0.0.1 B. 172.16.0.10 C. 15.1.5.6 D. 127.256.0.1

nielit-2018 computer-networks ip-addressing

Answer key

5.22.6 IP Addressing: NIELIT 2022 April Scientist B | Section B | Question: 116



The routing table of a router is shown below:

Destination	Subnet Mask	Interface
128.75.43.0	255.255.255.0	Eth0
128.75.43.0	255.255.255.128	Eth1
192.12.17.5	255.255.255.255	Eth3
default	Eth2	

On which interfaces will the router forward packets addressed to destinations 128.75.43.16 and 192.12.17.10 respectively?

- A. Eth1 and Eth2 B. Eth0 and Eth2 C. Eth0 and Eth3 D. Eth1 and Eth3

nielit2022apr-scientistb computer-networks ip-addressing subnetting

Answer key

5.22.7 IP Addressing: NIELIT Scientist B 2020 November: 117



You are working with a network that is 172.16.0.0 and would like to support 600 hosts per subnet. What subnet mask should you use?

- A. 255.255.192.0
- B. 255.255.224.0
- C. 255.255.252.0
- D. None of the options

nielit-scb-2020 subnetting computer-networks ip-addressing

Answer key

5.23

IP Packet (1)

5.23.1 IP Packet: NIELIT 2016 MAR Scientist B - Section C: 55



A packet switching network

- A. is free.
- B. can reduce the cost of using an information utility.
- C. allows communications channel to be shared among more than one user.
- D. both (B) and (C).

nielit2016mar-scientistb computer-networks ip-packet

Answer key

5.24

Information Theory (1)

5.24.1 Information Theory: NIELIT 2021 Dec Scientist A - Section B: 90



To simulate a analog signal of frequency f , bandwidth requirement of channel is :

- A. $2f$
- B. f
- C. $f/2$
- D. $f/4$

nielit2021dec-scientista information-theory computer-networks

Answer key

5.25

Logical Reasoning (2)

5.25.1 Logical Reasoning: NIELIT Scientist B 2020 November: 17



Directions: For the Assertion (A) and Reason (R) below, choose the correct alternative from the following:

- I. Both (A) and (R) are true and (R) is the correct explanation of (A)
- II. Both (A) and (R) are true and (R) is not the correct explanation of (A)
- III. (A) is true but (R) is false
- IV. (A) is false but (R) is true

Assertion (A):

Moon cannot be used as a satellite for communication

Reason (R)

Moon does not move in the equatorial plane of the Earth

- A. *I* B. *II* C. *III* D. *IV*

nielit-scb-2020 computer-networks logical-reasoning

5.25.2 Logical Reasoning: NIELIT Scientist B 2020 November: 50



In _____, other nodes verify the validity of the block by checking that the hash of the data of the block is less than a present number.

- A. Proof of Burn B. Proof of STAKE
C. Proof of Work D. All of the options

nielit-scb-2020 computer-networks logical-reasoning

Answer key

5.26

Network Addressing (6)

5.26.1 Network Addressing: NIELIT 2017 DEC Scientific Assistant A - Section B: 27

Which of the following is a class *B* host address ?



- A. 230.0.0.0 B. 130.4.5.6 C. 230.7.6.5 D. 30.4.5.6

nielit2017dec-assistanta computer-networks network-addressing ip-addressing

Answer key

5.26.2 Network Addressing: NIELIT 2017 DEC Scientific Assistant A - Section B: 28

There is a need to create a network that has 5 subnets, each with at least 16 hosts. Which one is used as classful subnet mask ?



- A. 255.255.255.192 B. 255.255.255.248
C. 255.255.255.240 D. 255.255.255.224

nielit2017dec-assistanta computer-networks network-addressing subnetting

Answer key

5.26.3 Network Addressing: NIELIT 2017 DEC Scientific Assistant A - Section B: 46

An Ethernet destination address 07 – 01 – 12 – 03 – 04 – 05 is :



- A. Unicast address B. Multicast address
C. Broadcast address D. All of the options

nielit2017dec-assistanta computer-networks network-addressing

Answer key

5.26.4 Network Addressing: NIELIT 2017 DEC Scientist B - Section B: 36



If the number of networks and number of hosts in class B are 2^m , $(2^n - 2)$ respectively. Then the relation between m, n is

- A. $3m = 2n$ B. $7m = 8n$ C. $8m = 7n$ D. $2m = 3n$

nielit2017dec-scientistb computer-networks network-addressing

Answer key

5.26.5 Network Addressing: NIELIT Scientist B 2020 November: 66



Which of the following can be used when creating a pool of global addresses instead of the netmask command?

- A. /(slash notation) B. prefix-length C. no mask D. block-size

nielit-scb-2020 computer-networks network-addressing

Answer key

5.26.6 Network Addressing: NIELIT Scientist B 2020 November: 74



In classful addressing, a large part of the available addresses are _____.

- A. Dispersed B. Blocked C. Wasted D. Reserved

nielit-scb-2020 computer-networks network-addressing

Answer key

5.27 Network Layer (5)

5.27.1 Network Layer: NIELIT 2016 MAR Scientist B - Section C: 54



Which of the following is not a transceiver function?

- A. Transmission and receipt of data B. Checking of line voltages
C. Addition and subtraction of headers D. Collision detection

nielit2016mar-scientistb computer-networks network-layer

Answer key

5.27.2 Network Layer: NIELIT 2021 Dec Scientist B - Section B: 62



A device that allows two or more computers on different networks to communicate with each other is _____.

- A. Hub B. Bridge C. Repeater D. Gateway

nielit-2021-it-dec-scientistb computer-networks network-layer

5.27.3 Network Layer: NIELIT Scientist B 2020 November: 108



_____ tells a firewall about how to reassemble a data stream that has been divided into packets.

- A. The source routing feature
- B. The number in the header's identification field
- C. The destination *IP* address
- D. The header checksum field in the packet header

nielit-scb-2020 computer-networks network-layer

Answer key 

5.27.4 Network Layer: NIELIT Scientist B 2020 November: 48



The _____ command will show you the translation table containing all the active *NAT* entries.

- A. show ip nat translations
- B. show ip nat tl
- C. show ip nat states
- D. none of the options

nielit-scb-2020 computer-networks network-layer

Answer key 

5.27.5 Network Layer: NIELIT Scientist B 2020 November: 64



In an *IPv6* header, the traffic class field is similar to the _____ field in the *IPv4* header.

- A. *TOS* field
- B. Fragmentation field
- C. Fast Switching
- D. Option field

nielit-scb-2020 computer-networks ip-addressing network-layer

Answer key 

5.28

Network Protocols (6)

5.28.1 Network Protocols: NIELIT 2016 DEC Scientist B (IT) - Section B: 31



A set of rules that governs data communication:

- A. Rule
- B. Medium
- C. Link
- D. Protocol

nielit2016dec-scientistb-it computer-networks network-protocols

Answer key 

5.28.2 Network Protocols: NIELIT 2017 DEC Scientific Assistant A - Section B: 32



Which NetWare protocol provides link - state routing ?

- A. *NLSP*
- B. *RIP*
- C. *SAP*
- D. *NCP*

nielit2017dec-assistanta computer-networks routing network-protocols

Answer key 

5.28.3 Network Protocols: NIELIT 2017 OCT Scientific Assistant A (CS) - Section B: 23



Which of the following would not be specified in a communication protocol?

- A. Header contents
- B. Trailer contents
- C. Error Checking
- D. Data content of message

nielit2017oct-assistanta-cs computer-networks network-protocols

Answer key 

5.28.4 Network Protocols: NIELIT 2018-53



Identify the true statement from the given statements.

1. HTTP may use different TCP connection for different objects of a webpage if non-persistent connections are used
2. FTP uses two TCP connections, one for data and another control
3. TELNET and FTP can only use TWO connections at a time

- A. 1
- B. 1 and 2
- C. 2 and 3
- D. 1, 2 and 3

nielit-2018 computer-networks network-protocols

Answer key 

5.28.5 Network Protocols: NIELIT Scientific Assistant A 2020 November: 44



Which among the following types of Server filters Website Traffic?

- A. POP Server
- B. Database Server
- C. Proxy Server
- D. Mail Server

nielit-sta-2020 computer-networks network-protocols

Answer key 

5.28.6 Network Protocols: NIELIT Scientific Assistant A 2020 November: 95



The router table contains addresses belonging to _____ protocol(s).

- A. a single
- B. two
- C. multiple
- D. none of the options

nielit-sta-2020 computer-networks network-protocols

Answer key 

5.29

Network Security (6)

5.29.1 Network Security: NIELIT 2016 MAR Scientist C - Section C: 38



How many bits are required to encode all twenty six letters, ten symbols, and ten numerals?

- A. 5
- B. 6
- C. 7
- D. 46

Answer key 

5.29.2 Network Security: NIELIT 2017 DEC Scientific Assistant A - Section B: 14

PGP encrypts data by using a block cipher called :



- A. international data encryption algorithm
- B. private data encryption algorithm
- C. internet data encryption algorithm
- D. none of the options

nielit2017dec-assistanta computer-networks network-security

Answer key 

5.29.3 Network Security: NIELIT 2017 DEC Scientific Assistant A - Section B: 49

Why is one - time password safe ?



- A. It is easy to generate
- B. It cannot be shared
- C. It is different for every access
- D. It can be easily decrypted

nielit2017dec-assistanta computer-networks network-security

Answer key 

5.29.4 Network Security: NIELIT 2021 Dec Scientist A - Section B: 101

The DoS attack, in which the attacker establishes a large number of half-open or fully open TCP connections at the target host is _____.



- A. Vulnerability attack
- B. Bandwidth flooding
- C. Connection flooding
- D. UDP flooding

nielit2021dec-scientista computer-networks tcp network-security

Answer key 

5.29.5 Network Security: NIELIT 2021 Dec Scientist B - Section B: 75

What is WPA?



- A. wi-fi protected access
- B. wired protected access
- C. wired process access
- D. wi-fi process access

nielit2021dec-scientistb network-security computer-networks

Answer key 

5.29.6 Network Security: NIELIT Scientist B 2020 November: 62

Data leakage threats are done by internal agents. Which of them is not an example of an internal data leakage threat?



- A. Data leak from stolen credentials from desk
- B. Data leak by partners
- C. Data leak by 3rd party apps
- D. All of the options

Answer key

5.30

Osi Model (5)

5.30.1 Osi Model: NIELIT 2021 Dec Scientist A - Section B: 66



Which NetWare protocol works on layer 3-network layer of the OSI model ?

- A. IPX B. NCP C. SPX D. NetBIOS

nielit2021dec-scientista computer-networks network-layer osi-model

Answer key

5.30.2 Osi Model: NIELIT 2021 Dec Scientist B - Section B: 84



TCP/IP model does not have _____ layer but OSI model have this layer.

- A. session layer B. transport layer
C. application layer D. network layer

nielit2021dec-scientistb computer-networks osi-model tcp

Answer key

5.30.3 Osi Model: NIELIT 2021 Dec Scientist B - Section B: 88



Each layer of the OSI model receives services or data from a _____ layer.

- A. below layer B. above layer
C. both (A) and (B) D. None of the above

nielit2021dec-scientistb osi-model computer-networks

Answer key

5.30.4 Osi Model: NIELIT 2021 Dec Scientist B - Section B: 89



Who developed standards for the OSI reference model ?

- A. ANSI – American National Standards Institute
B. ISO – International Standards Organization
C. IEEE – Institute of Electrical and Electronics Engineering
D. ACM – Association for Computing Machinery

nielit2021dec-scientistb computer-networks osi-model

Answer key

5.30.5 Osi Model: NIELIT 2021 Dec Scientist B - Section B: 98



Which is the layer that converts packets to Frames and Frames to packets in the

OSI model ?

- A. Physical Layer
- B. Data Link Layer
- C. Network Layer
- D. Transport Layer

nielit2021dec-scientistb osi-model data-link-layer computer-networks

Answer key 

5.31

Queuing Delay (1)

5.31.1 Queuing Delay: NIELIT 2021 Dec Scientist A - Section B: 75



In a network, If **P** is the only packet being transmitted and there was no earlier transmission, which of the following delays could be zero ?

- A. Propagation delay
- B. Queuing delay
- C. Transmission delay
- D. Processing delay

nielit2021dec-scientista computer-networks propagation-delay queuing-delay transmission-delay processing-delay

Answer key 

5.32

Segmentation (1)

5.32.1 Segmentation: NIELIT 2017 DEC Scientific Assistant A - Section B: 25



What is the value of acknowledgement field in a segment

- A. Number of previous bytes to receive
- B. Total number of bytes to receive
- C. Number of next bytes to receive
- D. Sequence of zero's and one's

nielit2017dec-assistanta computer-networks segmentation

Answer key 

5.33

Stop and Wait (1)

5.33.1 Stop and Wait: NIELIT 2022 April Scientist B | Section B | Question: 46



A sender uses the Stop-and-Wait **ARQ** protocol for reliable transmission of frames.

Frames are of size 1000 bytes and the transmission rate at the sender is 80 Kbps (1 Kbps = 1000 bits/second). Size of an acknowledgement is 100 bytes and the transmission rate at the receiver is 8 Kbps. The one-way propagation delay is 100 milliseconds. Assuming no frame is lost, the sender throughput is _____ bytes/second.

- A. 2500
- B. 2000
- C. 1500
- D. 500

nielit2022apr-scientistb computer-networks stop-and-wait normal

5.34

Subnetting (3)

5.34.1 Subnetting: NIELIT 2017 OCT Scientific Assistant A (IT) - Section B: 25



The address of a class B Host is to be split into subnets with a 3-bit subnet number. What is the maximum number of subnets and maximum number of hosts in each subnet?

- A. 8 subnets and 262142 hosts
- B. 6 subnets and 262142 hosts
- C. 6 subnets and 1022 hosts
- D. 8 subnets and 1024 hosts

nielit2017oct-assistanta-it computer-networks subnetting

Answer key 

5.34.2 Subnetting: NIELIT 2018-31



Identify the subnet mask for the given direct broadcast address of subnet is 201.15.16.31

- A. 255.255.192.192
- B. 255.255.255.198
- C. 255.255.255.240
- D. 255.255.255.240

nielit-2018 computer-networks subnetting

Answer key 

5.34.3 Subnetting: NIELIT Scientific Assistant A 2020 November: 102



You have a network ID of 192.168.10.0 and require at least 25 host IDs for each subnet, with the largest amount of subnets available. Which subnet mask should you assign?

- A. 255.255.255.192
- B. 255.255.255.224
- C. 255.255.255.240
- D. 255.255.255.248

nielit-sta-2020 computer-networks subnetting

Answer key 

5.35

TCP (3)

5.35.1 TCP: NIELIT 2016 MAR Scientist B - Section C: 53



In context of TCP/IP computer network models, which of the following is FALSE?

- A. Besides span of geographical area, the other major difference between LAN and WAN is that the later uses switching element.
- B. A repeater is used just to forward bits from one network to another one.
- C. IP layer is connected oriented layer in TCP/IP.
- D. A gateway is used to connect incompatible networks.

nielit2016mar-scientistb computer-networks tcp

Answer key 

5.35.2 TCP: NIELIT 2021 Dec Scientist A - Section B: 105



Which of the following system calls results in the sending of SYN packets ?

- A. Socket B. Bind C. Listen D. Connect

nielit2021dec-scientista operating-system tcp computer-networks

Answer key

5.35.3 TCP: NIELIT 2022 April Scientist B | Section B | Question: 71



Find out the maximum link speed at which a source can generate 1500 – byte TCP payloads with packet lifetime of upto 120 msec before the 32–bit sequence numbers wrap around? Take into account TCP, IPv4, and Ethernet header overheads while doing all calculations.

- A. 137 Mbps B. 256 Mbps C. 299 Mbps D. 512 Mbps

nielit2022apr-scientistb computer-networks tcp

5.36

Transmission Delay (1)

5.36.1 Transmission Delay: NIELIT 2022 April Scientist B | Section B | Question: 101



Consider two hosts X and Y, connected by a single direct link of rate 10^6 bits/sec. The distance between the two hosts is 10,000 km and the propagation speed along the link is 2×10^8 m/sec. Host X send a file of 50,000 bytes as one large message to host Y continuously. Let the transmission and propagation delays be p milliseconds and q milliseconds, respectively. Then the value of p and q are

- A. $p = 50$ and $q = 100$
B. $p = 50$ and $q = 400$
C. $p = 100$ and $q = 50$
D. $p = 400$ and $q = 50$

nielit2022apr-scientistb computer-networks propagation-delay transmission-delay analytical-aptitude

5.37

Transmission Media (1)

5.37.1 Transmission Media: NIELIT 2021 Dec Scientist B - Section B: 31



Which of the following is the efficient medium which can be used for higher bandwidth in broadband network?

- A. Coaxial cable B. Optical fiber C. CAT-V D. STP

nielit-2021-it-dec-scientistb computer-networks transmission-media

5.38

Transport Layer (3)

5.38.1 Transport Layer: NIELIT 2017 DEC Scientific Assistant A - Section B: 33

Which layer connects the network support layers and user support layers ?



- A. transport layer
- B. network layer
- C. data link layer
- D. session layer

nielit2017dec-assistanta computer-networks network-layer transport-layer

Answer key

5.38.2 Transport Layer: NIELIT 2018-45

_____ transport layer protocol is used to support electronic mail



- A. SNMP
- B. IP
- C. SMTP
- D. TCP

nielit-2018 computer-networks transport-layer

Answer key

5.38.3 Transport Layer: NIELIT 2021 Dec Scientist B - Section B: 92

Which address is used to identify a process on a host by the transport layer?



- A. physical address
- B. logical address
- C. port address
- D. specific address

nielit2021dec-scientistb computer-networks transport-layer

Answer key

5.39 Ttl (1)

5.39.1 Ttl: NIELIT 2018-44

Time to Live(TTL) field in the IP datagram is used _____



- A. to optimize throughput
- B. to prevent packet looping
- C. to reduce delays
- D. to prioritize packets

nielit-2018 computer-networks ttl

Answer key

5.40 UDP (2)

5.40.1 UDP: NIELIT 2017 DEC Scientific Assistant A - Section B: 1

Correct expression for UDP user datagram length is



- A. length of UDP = length of IP- length of IP header's
- B. length of UDP= length of UDP-length of UDP header's
- C. length of UDP = length of IP+length of IP header's
- D. length of UDP = length of UDP+length of UDP header's

nielit2017dec-assistanta computer-networks udp

Answer key

5.40.2 UDP: NIELIT 2022 April Scientist B | Section B | Question: 63



Host A sends a UDP datagram containing 8880 bytes of user data to host B over an Ethernet LAN. Ethernet frames may carry data up to 1500 bytes (i.e. MTU = 1500 bytes). Size of UDP header is 8 bytes and size of IP header is 20 bytes. There is no option field in IP header. How many total number of IP fragments will be transmitted and what will be the contents of offset field in the last fragment?

- A. 6 and 925 B. 6 and 7400 C. 7 and 1110 D. 7 and 8880

nielit2022apr-scientistb computer-networks udp ip-packet fragmentation ethernet

5.41

Wireless Lan (1)

5.41.1 Wireless Lan: NIELIT 2021 Dec Scientist B - Section B: 104



What is the access point (AP) in a wireless LAN ?

- A. device that allows wireless devices to connect to a wired network
B. wireless devices itself
C. both device that allows wireless devices to connect to a wired network and wireless devices itself
D. all the nodes in the network

nielit2021dec-scientistb computer-networks wireless-lan network

Answer key

Answer Keys

5.0.1	C	5.0.2	A	5.0.3	A	5.0.4	A	5.0.5	D
5.0.6	C	5.0.7	C	5.0.8	D	5.0.9	D	5.0.10	B
5.0.11	A	5.0.12	C	5.0.13	A	5.0.14	C	5.0.15	A
5.0.16	D	5.0.17	C	5.0.18	D	5.0.19	B	5.0.20	C
5.1.1	D	5.2.1	C	5.2.2	C	5.3.1	A	5.4.1	D
5.5.1	A	5.5.2	A	5.6.1	C	5.7.1	C	5.7.2	C
5.7.3	C	5.8.1	C	5.9.1	B	5.9.2	D	5.10.1	D
5.11.1	B	5.11.2	A	5.12.1	A	5.12.2	C	5.12.3	D
5.12.4	B	5.13.1	C	5.14.1	B	5.14.2	D	5.15.1	D
5.16.1	A	5.17.1	A	5.18.1	C	5.19.1	C	5.19.2	B

5.20.1	D
5.22.2	B
5.22.7	C
5.26.1	B
5.26.6	C
5.27.5	A
5.28.5	C
5.29.4	C
5.30.3	C
5.33.1	A
5.35.2	D
5.38.2	D
5.41.1	A

5.20.2	C
5.22.3	B
5.23.1	D
5.26.2	D
5.27.1	C
5.28.1	D
5.28.6	A
5.29.5	A
5.30.4	B
5.34.1	X
5.35.3	C
5.38.3	C

5.20.3	C
5.22.4	A
5.24.1	A
5.26.3	B
5.27.2	D
5.28.2	A
5.29.1	B
5.29.6	C
5.30.5	B
5.34.2	D
5.36.1	D
5.39.1	B

5.21.1	A
5.22.5	C
5.25.1	A
5.26.4	C
5.27.3	A
5.28.3	D
5.29.2	A
5.30.1	A
5.31.1	B
5.34.3	B
5.37.1	B
5.40.1	A

5.22.1	C
5.22.6	A
5.25.2	C
5.26.5	B
5.27.4	A
5.28.4	B
5.29.3	C
5.30.2	A
5.32.1	C
5.35.1	C
5.38.1	A
5.40.2	C